The Law of Graphic Equalisation — Total Perspective Vortex Grounding

Self-Model Dependence and the Cost of Cross-Frame Transition

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DRAFT — draft-2026-09-06.1 r3 (6 September 2026). This document is the *01g* grounding companion to *The Law of Graphic Equalisation — Theory*, *Mind and Self Grounding*, *Qualia and Consciousness Grounding*, and *The Law of Narrative Momentum*. It defines the Total Perspective Vortex as a recursive failure of a demand for total perspective. Its positive account is bounded: frames construct models, bridges, and costly transitions under support conditions. It neither makes a clinical classification nor treats a model of another frame as possession of that frame's experience.

Abstract

The phrase "Total Perspective Vortex" can sound as though it names an extraordinary mental state. This paper uses it instead as a defined formal term: the recursive failure produced by demanding a complete, frame-free total perspective from a finite frame. Graphic Equalisation already gives the reason a view is situated, and Mind and Self Grounding establishes that a self-model is constructed rather than complete. This paper does not reopen either result. It tracks their joint consequence for moving between perspectives. A frame F organises an item x as $P_F(x)$, carries a self-model $M_F(F)$, and can infer a model $M_F(K)$ of another frame K without thereby instantiating K's perspective. A mediating frame W can construct a bounded bridge $B^W_{F,K}$ and a representational transition $\tau^W_{F\to K}$. Such a transition is not a leap outside perspective: it is a new situated construction, assessed relative to support S_W.

Narrative Momentum supplies the cost form. Reusing carried organisation is the free continuation; altering it requires a specified intervention. Accordingly, this paper defines conditional transition cost $C_\tau(\tau \mid S_W)$ and a conditional self-model risk ordering. The latter increases, within a specified grounding and with relevant support held fixed, when a transition changes an organisation on which the frame's self-model is more dependent, central or coupled, less redundant, and less supported. It is a conceptual conditional quantity, not a clinical score, a universal numerical law, evidence that the transition is true, or a credential for any experience. The paper ends by specifying calibration tests and falsifiers for the bridge and cost claims.

1 Scope, dependency, and terminology guard

This is a focused Graphic Equalisation companion. It inherits the relational floor, the running graph, frame-relative grounding, and refusal of a view from nowhere from *Graphic Equalisation*; it inherits the structural incompleteness of self-models from *Mind and Self Grounding*; and it inherits the reuse-versus-alteration cost form from *Narrative Momentum*. It adds neither a consciousness theory nor a general proof that self-models are incomplete. Its narrower question is what a frame can construct when it tries to relate its perspective to another perspective without smuggling in a total view.

Terminological guard. *Total Perspective Vortex* (TPV) is a term defined by this paper. It is not attributed here to a literary, clinical, or scientific source. It names a failed demand and its recursive repair attempt, not a diagnosis, a disorder, a special capacity, or a credential. In particular, this paper does not claim that derealisation, disorientation, distress, or any other reported condition is required for, evidences, or authenticates the construction described here. Questions of clinical meaning, care, prevalence, and measurement are outside its scope.

Three distinctions keep the claim bounded. First, $M_F(K)$ is an inferred model of K *in* F; it is not direct possession of K's experience, a frame-free comparison class, or proof that F has reproduced every relation operative in K. Second, a transition cost is a cost of alteration under named conditions, not a measure of truth, value, courage, severity, or reality. Third, the formal order introduced below is conditional on a chosen grounding; it is not an empirical numerical law until a grounding supplies observables, units, and a measurement procedure.

The nearest established vocabulary is adjacent rather than identical. Predictive coding models treat perceptual processing as hierarchically organised prediction and residual error (Rao and Ballard, 1999; Friston, 2010); Metzinger's account makes self-model transparency a central constraint (Metzinger, 2003, 2020); and metacognition research studies measurable divergence between confidence and performance (Fleming, 2024). Those accounts concern particular theoretical or empirical machinery. The contribution here is the arrangement of frame-relative perspective, bridge construction, and conditional alteration cost. It makes no claim that any cited account establishes the present formalism.

2 Frames, aperture, and perspective

A frame is a locus at which distinctions are made. Its aperture is the set and grain of distinctions it can currently organise; it is not a hidden container of all distinctions it does not presently hold. Let x denote an item, event, or relation as individuated at a containing level.

Definition 2.1 (Frame-relative perspectival organisation)**.** *For a frame F,* frame-relative perspectival organisation *is*

$$P_F(x) = \mathrm{Org}_F(x; A_F, H_F, \Gamma_F),$$

where A_F is F's operative aperture, H_F its carried, currently operative history-bearing structure, and Γ_F its admissibility grammar. $P_F(x)$ is not an item copied unchanged into F; it is the organisation F can make and use of x under those conditions.

This notation does not turn perspective into an inner picture. It names an operative organisation: distinctions that are available, related, and consequential to F's continuation. The same x can support different $P_F(x)$ and $P_K(x)$ without either difference licensing a frame-free residual. A containing frame can compare relations that neither lower frame represents, but that comparison is $P_W(\cdot)$ and therefore carries W's aperture and grammar.

Remark 2.2 (Aperture is a boundary, not a defect)*.* A finite aperture neither makes a frame irrational nor prevents it from learning, modelling, or correcting. It fixes the condition under which a perspective is a perspective: F's organisation is made from a location with a finite operative vocabulary. An absent distinction is not thereby a value of zero, and an unavailable relation is not direct evidence that no frame can hold it.

3 Carried trajectory, self-model, and model of another

A frame carries a trajectory through its own running organisation. To use that trajectory as an object of reasoning, it constructs a representational model; that model is not identical with the carried course. This is the inherited self-model distinction, used here only as a dependency.

Definition 3.1 (Self-model and other-model)**.** *$M_F(F)$ is the model of F that F can construct, retain, and use under its operative aperture. For another frame K, $M_F(K)$ is F's inferred model of K: the organisation F uses to predict, interpret, coordinate with, or otherwise reason about K.*

The subscripts are load-bearing. $M_F(K)$ may incorporate reports, observed consequences, shared records, and a model of how K organises an item. It remains F's model. Replacing $M_F(K)$ with $P_K(x)$ erases the relation that needs explaining: F has constructed a representation *about* K, not stepped into a view from nowhere or acquired K's direct experience. Conversely, treating the model as worthless because it is not identity is another error: a model can be calibrated, useful, and revisable without being total.

Definition 3.2 (Carried trajectory)**.** *Let $T_F = \langle P_{F,t}(x)\rangle_{t\leq n}$ denote the selectively retained, currently operative succession of F's perspectival organisation. The carried trajectory is the part of T_F represented in present support and available to shape F's next organisation.*

The qualification "selectively retained" is essential. Narrative Momentum does not treat all past states as equally carried; it treats the history-bearing structure still operative in the present as what persists. Hysteresis is a close example: present response can retain selected branch or extremum information without retaining a total history (Mayergoyz, 1991, 2003). In an inferential setting, belief revision likewise studies how a change can preserve as much of an existing set as the specified revision permits (Alchourrón et al., 1985). Neither literature proves the present account; both identify the reuse and alteration structure that the account requires.

4 The Total Perspective Vortex and non-totalisation

The demand that begins the vortex is: "construct one perspective that is complete, frame-free, and total over all perspectives." A finite frame can respond by adding a model of itself, then a model of that model, then a containing observer of both. The recursion changes the represented object but never supplies the demanded location.

Definition 4.1 (Total Perspective Vortex)**.** *The* Total Perspective Vortex *is the recursive failure of a frame's demand for a complete, frame-free total perspective: every proposed totalising model is constructed in a finite frame, so its conditions of construction are relocated rather than eliminated; adding a model of those conditions restarts the same demand at the containing organisation.*

Proposition 4.2 (Non-totalisation)**.** *No finite frame obtains a complete, frame-free total perspective. A containing observer relocates the aperture and self-model dependency from F to the containing frame; it does not solve them.*

This is not a second proof of generic self-model incompleteness. It is the direct consequence of applying the inherited frame condition to the stronger demand for totality. If W represents F and K, W can construct relations unavailable to either. But its representation is $P_W(\cdot)$, conditioned by A_W, H_W, and Γ_W. Naming a still larger V repeats the relocation. The result does not forbid increasingly adequate models or multi-frame coordination. It forbids calling any finite, situated construction the completion of a frame-free total perspective.

Corollary 4.3 (No vortex is required for ordinary modelling)**.** *F need not demand totalisation in order to model K, revise a self-model, or construct a useful bridge. The vortex is a failure mode of an impossible demand, not the price of perspective-taking itself.*

5 Jumping off the tower: the cross-frame bridge

"Jumping off the tower" names the practical formal move: stop trying to complete an infinite tower of observers and instead specify the finite bridge a containing frame can actually construct. This is not an escape from frames. It is a change from totalisation to accountable mediation.

Definition 5.1 (Cross-frame bridge)**.** *Given frames F and K, a* cross-frame bridge *in mediating frame W is*

$$B^W_{F,K} = \langle E^W_{F,K},\ \Pi^W_{F,K},\ \Lambda^W_{F,K} \rangle,$$

where $E^W_{F,K} = \langle E^W_F, E^W_K \rangle$ *is the evidence, record, or contact relation W admits concerning F and K;* $\Pi^W_{F,K}$ *is W's correspondence or translation procedure; and* $\Lambda^W_{F,K}$ *states the bridge's loss, grain, and conditions of use. From its admitted evidence, W can construct* $M_W(F)$, $M_W(K)$, $\widehat{P}^W_F(x)$, *and* $\widehat{P}^W_K(x)$. *These W-indices are load-bearing: they are bounded constructions about existing frames, not F- or K-local organisation imported without mediation.*

A bridge can use shared language, behavioural regularities, measurements, reports, jointly constrained tasks, or formally specified mappings. Its quality is not guaranteed by its existence. The bridge must state what it preserves, what it leaves unrepresented, and which counterevidence would require revision. The model-of-another claim is therefore weaker and more useful than possession: it makes coordination and correction possible while retaining the difference between a construction and the perspective it models.

The derivation posits F and K as existing frames; it does not grant W exclusive, complete, or direct understanding of either. $P_F(x)$ remains the organisation operative in F, but it is not an input datum available to a W-mediated procedure. W works through E^W_F and the constructions it can make from that admitted relation. Another containing frame can construct a different or finer relation to F without any one representation exhausting F.

Definition 5.2 (Representational transition)**.** *A* representational transition *through W is a procedure*

$$\tau^W_{F\to K} : (E^W_F, M_W(F), B^W_{F,K}) \longmapsto \widehat{P}^W_K(x),$$

that produces a W-indexed, bridge-limited representation $\widehat{P}^W_K(x)$ *of how K might organise x. The arrow denotes constructed transformation, not transport of F into K or acquisition of K's experience.*

The transition is deliberately asymmetric. $\tau^W_{F\to K}$ begins with W's admitted relation to F and ends in a representation W can hold about K. A reverse construction has its own bridge, losses, and conditions. Equality, invertibility, or mutual transparency cannot be inferred from notation alone.

6 Transition cost and self-model risk

The bridge answers what can be constructed; it does not say what must be altered to use the construction. That question has the same shape as Narrative Momentum. A continuation that reuses the relevant carried structure is free with respect to that alteration; a continuation that requires the structure to be rewritten, bypassed, or made inoperative pays the specified alteration cost. Path-dependence and increasing-returns research provides familiar local instances in which historically selected structures become harder to abandon (Arthur, 1989; David, 1985; Pierson, 2000).

Definition 6.1 (Conditional transition cost)**.** *A* cost grounding $\mathfrak{C}$ *specifies a transition comparison class, a cost domain* $(\mathbb{C}_{\mathfrak{C}}, \preceq_{\mathfrak{C}})$*, its units or ordinal rule, and the support configuration* S_W *under which comparisons are made. The* conditional transition cost *is the value*

$$C^{\mathfrak{C}}_{\tau}(\tau^W_{F\to K} \mid S_W) \in \mathbb{C}_{\mathfrak{C}},$$

which $\mathfrak{C}$ assigns to constructing and maintaining the transition while preserving its stated conditions. It is undefined when no comparison class, cost order, or usable support condition has been supplied.

For a chosen transition class, let $H^\star$ be the carried organisation relevant to that class. Its reuse-versus-alteration form is

$$C^{\mathfrak{C}}_\tau(\tau \mid S_W) = C^{\mathfrak{C}}_{\text{reuse}}(\tau \mid S_W) \oplus \Delta^{\mathfrak{C}}_{\text{alter}}(H^\star, \tau \mid S_W), \qquad \Delta^{\mathfrak{C}}_{\text{alter}} \succeq_{\mathfrak{C}} 0.$$

Here $\Delta^{\mathfrak{C}}_{\text{alter}} = 0$ when τ reuses $H^\star$. Where τ requires $H^\star$ to be altered, bypassed, or made inoperative, $\Delta^{\mathfrak{C}}_{\text{alter}}$ is the specified alteration cost. This equality applies only when $\mathfrak{C}$ supplies an additive cost currency. Where it supplies only an ordinal comparison, the invariant claim is narrower: an otherwise matched transition that reuses $H^\star$ ranks no higher in alteration burden than one that requires its alteration. Neither form assumes a universal, monetary, conscious, or shared unit. In the free-continuation case, zero alteration increment in an additive implementation does not imply that total activity is costless in a containing frame.

Remark 6.2 (Reuse does not make a transition truth-preserving)*.* Reuse identifies the carried organisation that avoids further alteration cost in the selected class. It does not establish that the organisation is adequate, true, or normatively preferable. A bridge may reuse a badly calibrated model and still require revision on independent evidence.

Definition 6.3 (Conditional self-model risk ordering)**.** *A* risk grounding $\mathfrak{R}$ *specifies a comparison class $\mathcal{T}_{\mathfrak{R}}$, an ordered risk domain $(\mathbb{Q}_{\mathfrak{R}}, \preceq_{\mathfrak{R}})$, four ordered factor domains $(\mathbb{D}_{\mathfrak{R}}, \preceq_D)$, $(\mathbb{C}_{\mathfrak{R}}, \preceq_C)$, $(\mathbb{R}_{\mathfrak{R}}, \preceq_R)$, and $(\mathbb{S}_{\mathfrak{R}}, \preceq_S)$, and maps*

$$d, c, r, s : \mathcal{T}_{\mathfrak{R}} \longrightarrow \mathbb{D}_{\mathfrak{R}}, \mathbb{C}_{\mathfrak{R}}, \mathbb{R}_{\mathfrak{R}}, \mathbb{S}_{\mathfrak{R}},$$

respectively measuring self-model dependence, centrality or coupling, redundancy, and post-transition remaining support. Each factor order is oriented so that larger elements mean more of the named factor. The initial support configuration is S^0_W; s_τ is what remains after the particular transition and is therefore not held fixed by conditioning on S^0_W. The risk map is typed as an order-preserving map

$$\Phi_{\mathfrak{R}} : \mathbb{D}_{\mathfrak{R}} \times \mathbb{C}_{\mathfrak{R}} \times \mathbb{R}^{\text{op}}_{\mathfrak{R}} \times \mathbb{S}^{\text{op}}_{\mathfrak{R}} \longrightarrow \mathbb{Q}_{\mathfrak{R}},$$

and defines

$$\mathcal{R}^{W,\mathfrak{R}}_F(\tau \mid S^0_W) = \Phi_{\mathfrak{R}}(d_\tau, c_\tau, r_\tau, s_\tau).$$

This orders vulnerability to loss of calibration, continuity, or bridge use under the selected grounding; it is not a clinical score or a universal numerical law.

Hypothesis 6.4 (Conditional dependence risk)**.** *For $\tau_1, \tau_2 \in \mathcal{T}_{\mathfrak{R}}$ with the same target, bridge-quality criterion, initial support S^0_W, and measurement grounding, if $d_{\tau_1} \succeq_D d_{\tau_2}$, $c_{\tau_1} \succeq_C c_{\tau_2}$, $r_{\tau_1} \preceq_R r_{\tau_2}$, and $s_{\tau_1} \preceq_S s_{\tau_2}$, with at least one strict inequality, then*

$$\mathcal{R}^{W,\mathfrak{R}}_F(\tau_2 \mid S^0_W) \prec_{\mathfrak{R}} \mathcal{R}^{W,\mathfrak{R}}_F(\tau_1 \mid S^0_W).$$

An intervention may add initial support or redundancy, thereby defining a new comparison configuration; it cannot be described as holding S^0_W fixed. Neither a higher risk order nor a higher transition cost establishes the truth, value, or clinical meaning of the transition.

The conditionals do the work. The order applies only after a risk grounding supplies its comparison class, observables, and ordering rule. A false model can be expensive to alter and a true correction can be cheap when support and redundancy are strong. Conversely, a low cost neither proves safety nor truth. The claim is that the named configuration determines why formally similar transitions can have different alteration burdens, and that this claim can break at a specified measurement boundary.

7 Failure modes, calibration, tests, and falsifiers

The account fails if its limits are silently removed. Four recurrent mistakes provide direct calibration checks.

1. *Totalisation error:* treat a containing model as a frame-free view. The repair is to index the construction to W and state A_W, Γ_W, and its bridge conditions.

2. *Possession error:* treat $M_W(K)$ or $\widehat{P}_K^W(x)$ as K's direct experience. The repair is to preserve the model, bridge, and target indices and list unrepresented relations.

3. *Cost-as-truth error:* infer that difficulty authenticates the transition. The repair is an independent bridge calibration: compare its predictions or coordination outcomes with admitted evidence, not with its cost.

4. *Clinical reification error:* promote the formal variables into a diagnosis, a severity ranking, or a required phenomenology. The repair is to keep them as conditional structural quantities until an appropriate empirical and ethical measurement programme independently licenses more.

Calibration protocol. For a specified application, first declare a cost grounding $\mathfrak{C}$ and risk grounding $\mathfrak{R}$: their comparison classes, units or ordering rules, initial support S_W^0, and independently observable measures for $d_\tau, c_\tau, r_\tau, s_\tau$. Then construct at least two bridges with their losses stated. A bridge earns use only insofar as it improves specified prediction, coordination, compression, or error detection relative to an available comparison; it is not validated by eloquence or by an assertion of empathy. Compare transitions within the declared class. An intervention that adds redundancy or support creates a new initial configuration; the hypothesis predicts lower ordered risk only when the added relation independently preserves or replaces the relevant support, not when it is merely additional description.

Proposition 7.1 (Falsifiers)**.** *The framework's claims are exposed at three levels. The non-totalisation result fails if a finite construction can supply a complete, frame-free perspective while retaining no unrepresented construction conditions. The bridge account fails if a purportedly unmediated transition can be shown to contain neither admitted evidence, translation, nor loss conditions while still relating non-identical apertures. The conditional-risk hypothesis fails in a grounded measurement programme if, within its declared comparison class and common initial support S_W^0, independently measured increases in dependence or coupling, decreases in redundancy, and decreases in post-transition remaining support fail to order risk as specified across the relevant cases.*

The last exposure is deliberately not a universal prediction. A negative result in one implementation can locate a bad operationalisation of d, c, r, s, a missing support variable, or a failure of the hypothesised ordering in that class. It cannot be repaired by declaring cost to be truth or by relabelling an ordinary disagreement as a vortex.

8 Prior art, relation to 01b/01f/03, and conclusion

This paper's relation to its neighbouring documents is division of labour. *Mind and Self Grounding* establishes that a self-model is constructed under a finite aperture and distinguishes calibration from completeness; 01g uses that result as the source of self-model dependence but does not repeat its generic proof. *Qualia and Consciousness Grounding* concerns whether determinate phenomenal organisation requires a second ontological kind; 01g does not reopen that question and does not infer experience from bridge construction. *Narrative Momentum*

supplies the present-structure and reuse-versus-alteration form; 01g applies it specifically to cross-frame representational transition rather than claiming a new law of history.

The adjacent research record supplies pieces, not a hidden proof. Predictive coding supplies models of hierarchical inference and residual updating (Rao and Ballard, 1999; Friston, 2010). Metzinger develops a substantial account of self-model transparency (Metzinger, 2003, 2020); Fleming reviews measurable calibration in metacognitive confidence (Fleming, 2024); hysteresis, path dependence, and AGM revision provide established analyses of retained state, lock-in, and conservative change (Mayergoyz, 1991, 2003; Arthur, 1989; David, 1985; Pierson, 2000; Alchourrón et al., 1985). None establishes a complete perspective, and none is claimed to validate a clinical use of TPV.

The demystification is exact. The Total Perspective Vortex is not a hidden realm behind ordinary perspective-taking. It is what occurs when a finite frame demands a view that would have to erase the very aperture and construction through which it is made. The alternative is not resignation or solipsism. It is finite, inspectable bridge construction: model another frame as another frame; state the bridge and its losses; reuse what carries forward where that is appropriate; and name the conditional cost of altering what the self-model depends on. A containing observer can improve the bridge, but it relocates rather than abolishes the conditions of perspective. That boundary is the result.

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